# Day 03: Inference for SLR

## Data for today

We're returning to the NHANES sitting time and depression risk score data. We're analyzing the relationship between depression risk score and sitting time.

> **Theoretical Model:**
>
> $(x_i, Y_i)$ are the data points; _____ and _____ are (unknowable) population parameters
> _____ $= \beta_0 + \beta_1 x_i + \varepsilon_i$ where $\varepsilon_i \overset{iid}{\sim} N(0, \text{_____})$

> **Fitted Model:**
>
> $\hat{Y}_i$ are predictions; _____ and _____ are estimated from the sample
> _____ $= \hat{\beta}_0 + \hat{\beta}_1 x_i$

```
summary(nhanes_lm)
```

```
Call:
lm(formula = dep_score ~ sedentary_hours, data = nhanes)

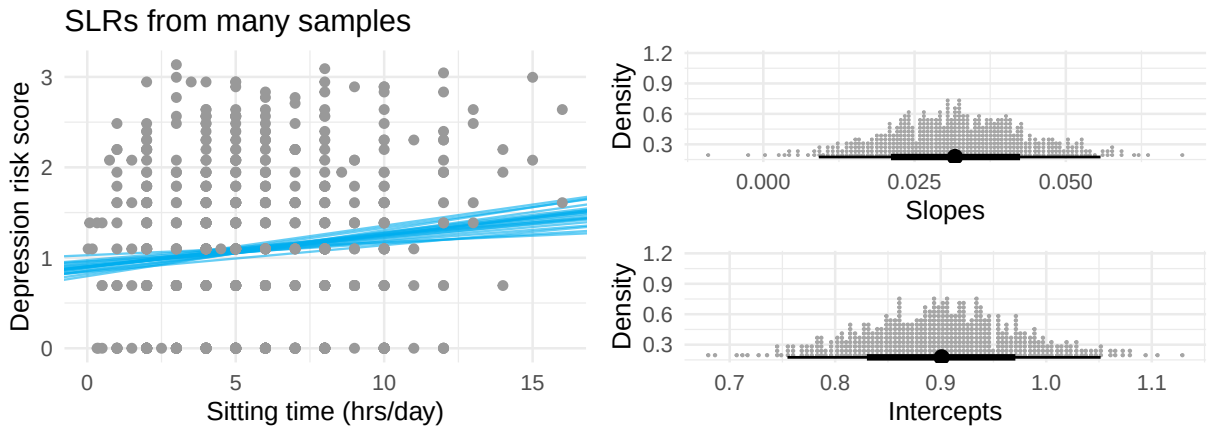
Residuals:
    Min      1Q  Median      3Q     Max
-1.2812 -0.9671  0.0059  0.6424  2.1370

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.90426    0.07359   12.29   <2e-16 ***
sedentary_hours  0.03141    0.01155    2.72   0.0067 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8835 on 657 degrees of freedom
Multiple R-squared:  0.01113,   Adjusted R-squared:  0.009629
F-statistic: 7.398 on 1 and 657 DF,  p-value: 0.006704
```

## Sampling variability

Every new sample gives us a slightly different fitted line.




> **Sampling Distribution**
>
> If we obtained many different random samples, what would the distribution of their _____ ($\hat{\beta}_0$ and $\hat{\beta}_1$) look like?

We use the **t distribution** to model the sampling distribution of a regression coefficient.

- Degrees of freedom: $df =$ _____
- Mean: $= \hat{\beta}_i$
- Standard deviation: $SE(\hat{\beta}_i) = \dfrac{\hat{\sigma}}{\sqrt{\sum(x_i - \overline{x})^2}}$

## Confidence interval

General Form:

$$\text{statistic} \pm \text{critical value} \times \text{standard error}$$

Form for regression coefficient:

$$\hat{\beta}_i \pm t^* \cdot SE(\hat{\beta}_i)$$

> **Critical value**
>
> $t^*$ is the value that cuts off the right tail area that matches your confidence level. For a 95% CI, find $t^*$ so that the two-sided tail area is 0.025 on each side, using `qt(.025, df = n-2)`.

### Hypothesis test

$$H_0 : \beta_i = \text{null value} \quad \text{vs.} \quad H_a : \beta_i \begin{cases} \neq \\ < \\ > \end{cases} \text{null value}$$

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{SE} = \frac{\hat{\beta}_i - \text{null value}}{SE(\hat{\beta}_i)}$$

Then find the appropriate tail area(s) under the t-distribution with $df = n - 2$.

```
tidy(nhanes_lm)
```

```
# A tibble: 2 x 5
  term            estimate std.error statistic  p.value
  <chr>              <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)       0.904     0.0736     12.3  2.18e-31
2 sedentary_hours   0.0314    0.0115      2.72 6.70e- 3
```

> **Try it**
>
> Verify the test statistic for $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$ using the table above.

**Example:** 90% confidence interval for $\beta_1$

$$\hat{\beta}_i \pm t^* \cdot SE(\hat{\beta}_i)$$

$t^* =$

```
qt(0.95, df = 659 - 2)
```

```
[1] 1.647176
```

**Interpretation:**

General template (reusable for any CI)

- We are _____% confident that for every 1 _____ increase in _____,
- _____ also _____
- by between _____ and _____

**Discussion:** Recall from Day01: we found no significant difference between the “High” and “Low” sitting groups. How does this result compare? Which do you trust more?